

Sample solution to Assignment (II)

Problem 1 (30 points)

Prove the following the characterizations of extraneous attributes.

1. $A \in X$ is *extraneous* in $X \rightarrow Y$ iff $Y \subseteq (X \setminus \{A\})_F^+$.

$$(\Leftarrow) Y \subseteq (X \setminus \{A\})_F^+ \iff F \models X \setminus \{A\} \rightarrow Y$$

We have $F \models F \setminus \{X \rightarrow Y\}$, so $Y \subseteq (X \setminus \{A\})_F^+ \iff F \models (X \setminus \{A\} \rightarrow Y) \cup F \setminus \{X \rightarrow Y\}$.

($\Rightarrow$) Suppose $A \in X$ is *extraneous* in $X \rightarrow Y$, so $X \setminus \{A\} \rightarrow Y$. So $Y \subseteq (X \setminus \{A\})_F^+$.

2. $B \in Y$ is *extraneous* in $X \rightarrow Y$ iff $B \in X_{F'}^+$, where $F' = (F \setminus \{X \rightarrow Y\}) \cup \{X \rightarrow (Y \setminus \{B\})\}$.

We have $B \in X_{F'}^+ \iff F' \models X \rightarrow B$, and $F' \models X \rightarrow (Y \setminus \{B\})$, so we have

$$F' \models X \rightarrow B \iff F' \models X \rightarrow Y.$$

Since $F' \models F \setminus \{X \rightarrow Y\}$, so $B \in X_{F'}^+ \iff F' \models F$.

Problem 2 (30 points)

Suppose that the following FD's hold on the relation schema $R(A, B, C, D)$

$$AB \rightarrow C, \quad AB \rightarrow D, \quad C \rightarrow A, \quad D \rightarrow B.$$

(1). List **all** candidate keys of R .

AB, CD, AD, BC .

(2). Show that R is not in BCNF and give a BCNF decomposition of R .

Since $C \rightarrow A \in F$ but C is not a superkey of R .

$\{A, C\}, \{B, D\}, \{C, D\}$.

(3). Is your BCNF decomposition dependency preserving? Explain your answer.

No, for $\{C \rightarrow A, D \rightarrow B\}^+$ is not equal to F^+ .

Problem 3 (30 points)

Let F be a set of FD's holds on a schema R and $R_1, \dots, R_n$ be a decomposition of R .

Furthermore, assume the following:

1. For every $X \rightarrow Y$ in F , there exists some R_i that contains all the attributes in XY .
2. At least one schema in the decomposition contains a candidate key of R .

Prove that the decomposition $R_1, \dots, R_n$ is join lossless. With this property, we can show that any decomposition produced by the 3NF synthesis algorithm is join lossless.

proof.

Let I be a relation instance of R . We prove that $I = \Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I)$.

First, we prove that

$$I \subseteq \Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I)$$

Let t be a tuple in I , then there are tuples $t_i \in \Pi_{R_i}(I)$ for $i = 1, \dots, n$ such that

$$t_1 = \Pi_{R_1}(t), t_2 = \Pi_{R_2}(t), \dots, t_n = \Pi_{R_n}(t).$$

Observe that

$$t = t_1 \bowtie t_2 \bowtie \dots \bowtie t_n \in \Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I).$$

That is, t is also in $\Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I)$.

Thus, we prove that $I \subseteq \Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I)$.

Then, we prove that

$$\Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I) \subseteq I$$

Let t be a tuple in $\Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I)$. Then there are $t_1, t_2, \dots, t_n$ in I such that

$$\Pi_{R_1}(t) = \Pi_{R_1}(t_1), \Pi_{R_2}(t) = \Pi_{R_2}(t_2), \dots, \Pi_{R_n}(t) = \Pi_{R_n}(t_n).$$

Assume w.l.o.g. that R_1 contains a candidate key of R . We next show that $t_1 = t$. This will conclude the proof since it implies that t is also in I .

Denote by X the set of attributes in R_1 and by Y the set of attributes in R but not in R_1 . Since X contains a candidate key of R , X^+ includes all the attributes of Y .

It suffices to show that t_1 and t agree on the attributes of Y since $\Pi_{R_1}(t_1) = \Pi_{R_1}(t)$.

To this end, let us consider the computation process of X^+ . In each iteration, a non-trivial FD $X' \rightarrow Y'$ is chosen and the attributes in $Y' \cap Y$ are added to the closure. By assumption, there exists some R_i containing all the attributes in $X'Y'$. By $X' \rightarrow Y'$ and $\Pi_{R_i}(t) = \Pi_{R_i}(t_i)$, we know that $\Pi_{Y \cap Y'}(t) = \Pi_{Y \cap Y'}(t_i) = \Pi_{Y \cap Y'}(t_1)$. By repeating this argument, we can see that t and t_1 agree on all the attributes of Y , $\Pi_Y(t) = \Pi_Y(t_1)$.

And since $\Pi_X(t) = \Pi_X(t_1)$, we have $\Pi_R(t) = \Pi_R(t_1)$, $t = t_1$.

Thus we prove that $\Pi_{R_1}(I) \bowtie \Pi_{R_2}(I) \bowtie \dots \bowtie \Pi_{R_n}(I) \subseteq I$

Problem 4 (10 points)

How long does it take you to finish the assignment? Give a score (1,2,3,4,5) to the difficulty of each problem. List all your collaborators if you have any.

As long as you answer it, you can get 10 points directly.